

**Determining the ideal G force level in order to optimize growth rate and length of
Raphanus Sativus.**

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Determining the ideal G force level in order to optimize growth rate and length of *Raphanus Sativus*.

An experiment was conducted in order to determine the G force which most optimized the growth rate and length of radish (*Raphanus Sativus*) seeds. Two groups of seeds were subjected to varied G forces, with one at 1G (Normal earth gravity or 9.8m/s) and 6Gs (58.8m/s) respectively. Radish seeds were selected randomly and put in groups of three for each test tube along with 5 ml of distilled water; 8 test tubes composed a trial run for both levels of G force being tested. Test tubes to be subjected to six Gs were inserted into a centrifuge, where the appropriate rpms to reach six Gs of force was set. Seeds to be exposed to one G were placed on a test tube holding rack and left unattended. After 24 hours the seeds were removed from each test tube, placed onto a paper towel sheet, their lengths were measured by caliper and individually entered into a trial spreadsheet, each group of seeds were returned to their original test tube and filled with another 5 ml of distilled water. This measuring procedure was repeated after another 24 hours, where after the seeds would be discarded. This procedure was repeated again twice, resulting in 72 trials.

A two sample t test was conducted in order to determine which gravity was most optimal for seed growth. After 24 hours p value of seeds subjected to 6 Gs of force yielded a result of 0.0046 much less than the alpha value of 0.05, this value was even smaller after 24 hours resulting in a value of 3.0×10^{-6} . These results lead to the conclusion that seeds subjected to higher G force grew more efficiently, this was also supported by the longer lengths of seeds in said situation.

Results gained from the experiment allow for a general prediction of how seed growth could be affected under conditions where they are subjected to G force higher than that of Earth's

(1 G). This allows for an understanding of how more efficient methods for plant growth can be achieved, as well as a general prediction of how plant seeds grow in extraterrestrial conditions.

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Introduction

Space travel is the next frontier in research and exploration. It is a place humans only have a limited knowledge on, and there is so much still unknown. Getting to space and sustaining life is a very challenging thing so, in order to meet this challenge, the basic life factors need to be met. There are many factors that come into play to ensure that life is sustained, is in check and occurs. Countless factors had to align to support the production of space, mass, and also life. There are effects that have an impact on growth and the sustainment of life. And they also interact with each other. One of these factors is gravity. Gravity is a force that applies itself on all things that contain mass, and have a crucial impact on the way the world works. It affects our lives through the way we travel, the way we live, and the things we do. Gravity can be defined as the universal force of attraction between all matter. Gravity simulates a “pulling” force between objects. Every planet has its own unique gravitational pull, and becomes part of the environment. Earth's acceleration of gravity is 9.8m/s^2 , or 1G. This is what keeps us stuck to the ground, keeps things on the floor when we place them down, and also holds plants in the ground when we plant them.

Many things can affect the way that plants grow, as plants all have an environment that can create the best growth, and this is usually what they are used to. This is the goal of what was being investigated in this research. What effect, if any, does a different force of gravity have on the growth of a plant?

Gravity is always in effect, and provides a restraining for growth as growing forces have to resist the force of gravity. Essentially, when there is less gravity, it should be easier to grow taller, and when there is more gravity, things should be shorter. This has been shown through

humans and space travel, as it has been found that astronauts grow taller when they reside in space, then return to their normal height after returning to earth for a while. This is because the human body is used to growing at a specific rate on earth when having to resist 1G of gravity, but when that disappears in space, those restraints no longer exist, and people become taller. This can be related to plants, as both have to grow under the conditions and restraints of gravity, so they would likely have to adapt to a different force of gravity if confronted with it. Previous research has shown that at a molecular level, the plants adapt towards their provided gravity. It was shown that under hypergravity, they would grow more horizontally than vertically, as gravity wouldn't stunt their growth vertically as much. This meant that the hypocotyl, or stem, would grow thicker and bend to a side. The opposite of this was true for microgravity, where the plants grew longer and had thinner hypocotyls. Further discussion on previous research occurs in the review of literature. This is crucial to know as it can be used for space travel. In space, there is no force of gravity acting on objects, which creates a microgravity condition. In contrast, on some other planets, if they have a larger mass, they will create an impact of more gravity. This would help clarify whether plants need to be grown in 1G, or any Gs. This could be utilized in space stations and on planets by growing plants on a centrifuge to simulate the optimal gravity load for growth.

The objectives of the study are to identify whether different forces of gravity had an impact on growth, which can be supported with some certainty from tests and statistical data. In addition to this, an objective is to collect data that is precise and accurate and can be beneficial in areas related to space travel and biology. Accuracy can be attained by keeping all seeds under testing in the closest possible environment besides the one being tested, and trying to replicate each trial exactly. Although the proper precautions were taken, it is impossible to eliminate lurking variables, which may still exist.

Seeds were selected and placed inside test tubes which were either kept under normal earth gravity by being placed in a rack or they were under hypergravity by having them placed in a centrifuge that simulated a 6G outward force. From there, the seeds were removed and compared to each other to determine any effects among the seeds.

This can be applied in the aeronautics industry and for space travel. The results can crucially identify whether plants need to be grown in a special gravity environment when in space, or if they can just be grown traditionally.

Review of Literature

This experiment conducted a comparison of the growth of radish plants at the normal gravitational level compared to the growth at an increased gravitational level. The purpose of the experiment was to determine whether plants could survive or possibly thrive under increased G's. The experiments simulated the gravity that was experienced on other planets to determine if life can be sustained. The measure of a G can be used to measure how much force these plants experienced. The G represents the acceleration of gravity compared to earth's Gravity of 9.8. This generally just represents the pull of gravity and can be simulated in multiple ways. Earth has a gravity of 9.8m/s and this is referenced to 1 G. In space and on other planets however, objects experience different G levels that can be higher than that on earth or zero, like in space.

Gravity can be defined as the force of attraction between two or more objects and includes all matter. Gravity is the force that provides things with weight, where the gravity is proportional to their true mass (Nordtvedt). Essentially, what this means is that any object with mass has its own gravitational pull. The more mass objects have, the more gravitational pull they have. This is why everyday objects on earth don't seem to attract each other, but planets and stars in space have heavy gravitational pulls.

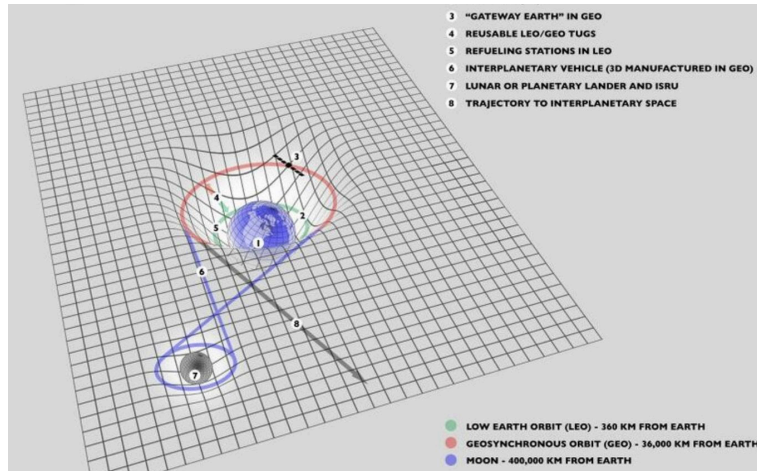


Figure 1. Gravity Diagram Vidmar, Matjaz. "Gravity Field Diagram"

Figure 1 above shows a gravity field diagram to represent visually how gravity works. This shows how objects and especially with large masses "curves the fabric" of space time to create a pull to any objects that enter that area (NASA). In this example, earth is shown in the middle, with the moon on the outside, and the object labeled 3 is a satellite. As shown, the satellite is within the gravitational pull of earth as it is under the bend of space time. Due to this reason, it will be pulled toward the earth.

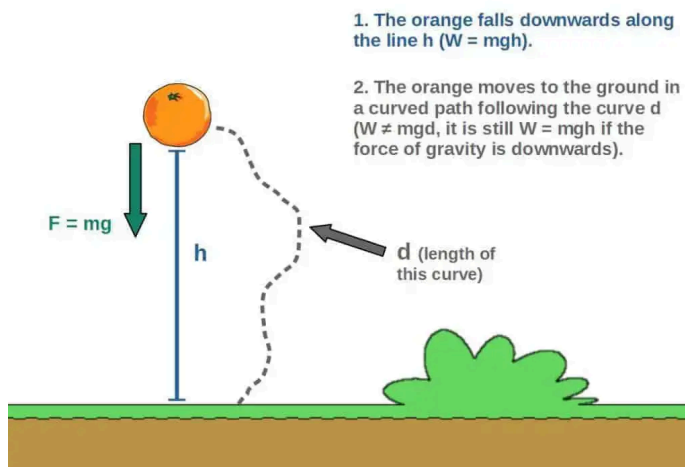


Figure 2. Representation of Gravity Hirvonen, Ville. "Does Gravity Do Work?"

Figure 2 shows the diagram for an example of how gravity works. This shows how when an object is dropped from space it falls downward. This shows that the force of gravity can be calculated by multiplying mg , or the mass of the object by the acceleration of gravity.

The conditions of gravity have effects on all sorts of organisms. Gravity iOS constantly affects earth and the organisms around it, so it has helped form how plants and organisms grow under earth's gravity (Narayanan). Through evolution, plants are adept and used to growing under the conditions of earth's gravity. Plants that undergo photosynthesis typically grow in an upwards direction, and they have to grow against the impact of earth's gravity. Changing the gravity level that a plant is under changes its growth drastically. A plant would have to adapt to the new gravity it is under and would therefore grow differently leading to different structure and functions.

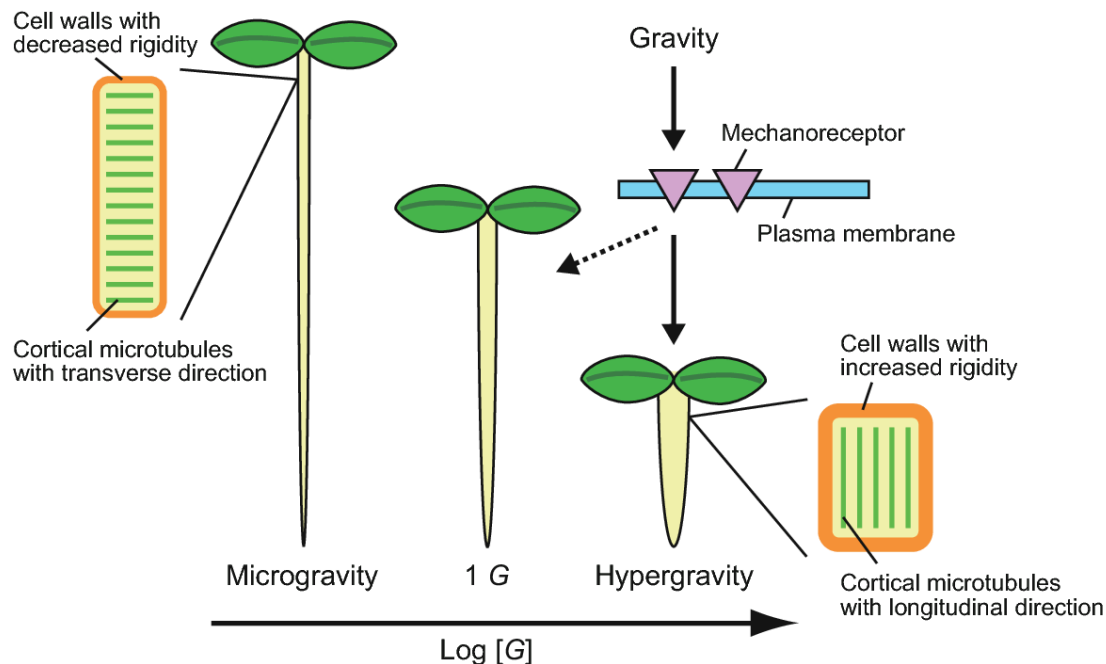


Figure 3. Plants vs Gravity Levels Soga, K. "Gravity Resistance in Plants"

Figure 3 above shows how plants typically adapt to differing G levels and conditions.

Under microgravity, plants have less weight and less resistance to grow vertically, so plants will be able to grow longer and have thinner diameter as less structure is required to keep the plants up. Under Hypergravity, Plants have more resistance and will therefore grow shorter with thicker diameters and have stover cell walls to keep them structurally sound under higher gravity with more weight.

Gravity is not something that can easily be manipulated, but it is not impossible. There is no way to outright change the gravity some object is experiencing, but it is possible to simulate gravity using a specific strategy-Gravity from acceleration. When someone gets in a car and starts driving, their body wants to stay stable, but the car wants the body to go forward as well, so these two opposing movements create a force on the person. This force is G force. One way to simulate this constantly without much work is using a centrifuge. This spins an object in a circle and creates artificial gravity through the opposing movement and centripetal force. The object doesn't want to go in an elliptical motion and wants to go straight, so if something is holding the object in, this creates force on the object, and can be controlled based on the speed of the object.



Figure 4. Centrifuge Experiment Lea, Robert. “Artificial Gravity”

Figure 4 above shows an experiment that simulated gravity. The mechanism is a centrifuge that had a person strapped down to it and would then rotate around. The outward force from the spinning created a gravitational force of 1G, simulating the gravity on Earth (Lea). This was intended for use in space so people could experience the normal gravity on Earth.

The topic of plant growth was vital in this experiment. Several factors all affect the growth of plants, though there are factors that more greatly facilitate the growth and survival of plants than others.

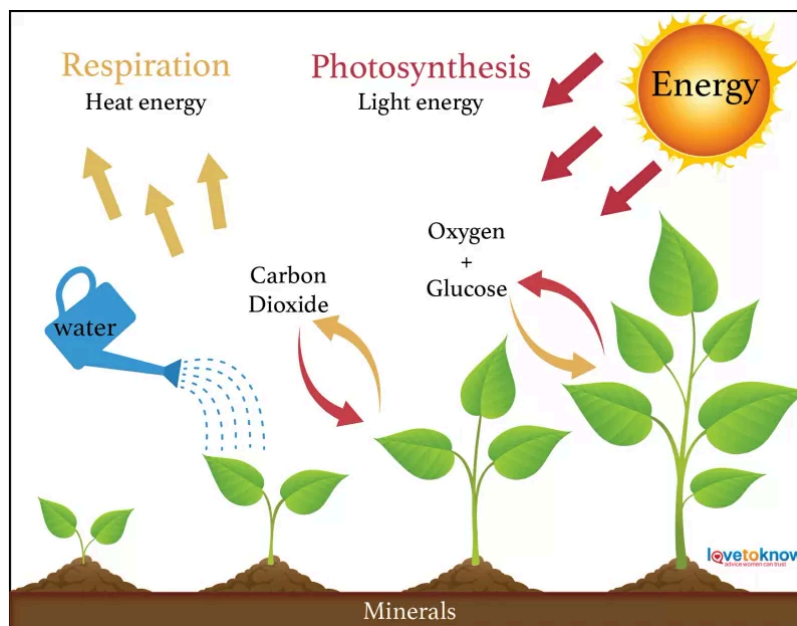


Figure 5. Factors Affecting Plant Growth Jeanne, Grunert. “4 Plant Growth Factors That Affect All Plants”

The figure above depicts four main factors that affect plant growth, those being water, light, temperature and nutrients, the image also shows the use of carbon dioxide which is vital for the process of photosynthesis. Water allows for minerals and critical elements to be dissolved in soil, these nutrients are then able to be transported into the plants cells through absorption of

water through the roots. Light allows for the process of photosynthesis in conjunction with water and carbon dioxide. Temperature encourages germination and growth of plants, as warmth triggers chemical reactions within plant cells which speed up processes of photosynthesis transpiration and respiration. Finally, absorption of nutrients facilitates the growth of plants as it makes plants less susceptible to viral and bacterial infections, as well as allowing for processes that do not stunt or slow plant growth.

The factors that were detailed in the figure above are all extremely important to plant growth, all working to allow plant cells to replicate and processes between cells/ the plants inner workings to function at their utmost efficiency. Though one factor detailed arguably holds the most importance, that being photosynthesis.

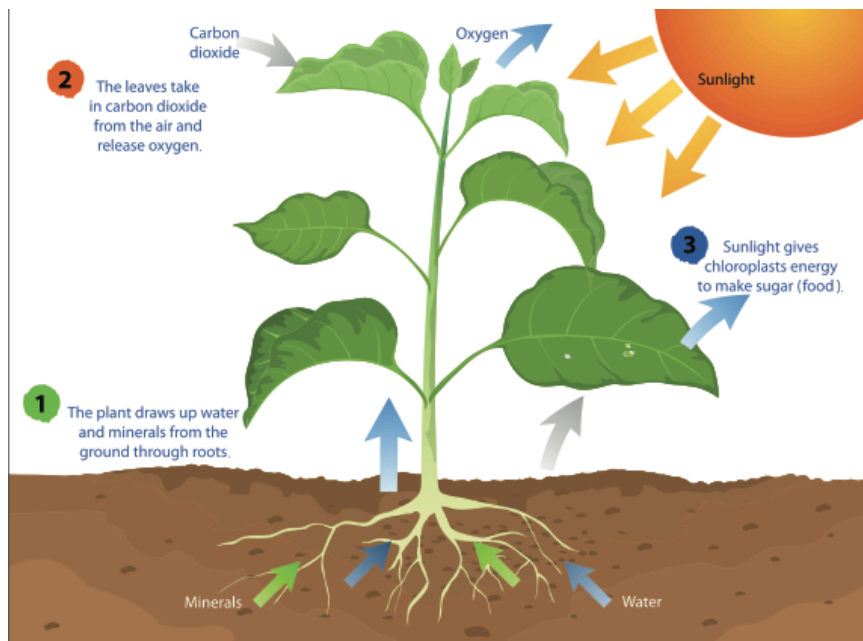


Figure 6. Photosynthesis Process Science and Technology Concepts Middle School. "What is Photosynthesis"

The figure above shows the process by which photosynthesis takes place in a step by step manner. The figure details how water is drawn through the roots of the plant, and how carbon dioxide is obtained through the leaves, which then is used to create sugar or food which the chloroplast creates with the use of sunlight (also absorbed by leaves).

This process is vital to the growth and development of a plant as it is necessary to create food for the plant which is energy which facilitates its growth. This is why the three components of photosynthesis (detailed in the figure above), being carbon dioxide, water (H_2O), and light, are integral to plant growth. Although the parts that allow for photosynthesis to occur must always be present, the environment which facilitates plant growth is variable.



Figure 7. Plant Growth Through Use of Hydroponic Technique Dr. Raymond, Wheeler.

“Growing Plants in Space”

The figure above shows Dr. Raymond Wheeler examining the growth of radishes which were being grown using a hydroponic technique. The image above depicts an environment in which plants are grown without soil, but in a water based nutrient mineral solution. The use of this technique by Wheeler was intended to explore methods that would allow for optimal growth

of plants in areas with different conditions than that of Earth, mainly being in reference to methods that would allow for the best growth in space.

The use of hydroponics sees great success with the growth of plants as the solution which the plant is grown in carries nutrients and minerals which aid greatly in a plants growth. The process also only uses about 10% of water that you would need for a plant in soil. This example of other avenues that facilitate plant growth represent the variability in plant growth that can be caused by many factors which affect said growth. The widespread use and popularity of hydroponics has been shown to yield great results, such as the image above. As well as being a more efficient use of resources and space.

Problem Statement

Problem statement:

To determine the ideal G force level in order to optimize growth rate and length of radishes.

Hypothesis:

A radish seed that was under normal 1G gravity conditions with regular watering will yield the best overall growth and growth rate.

Data Measured:

The independent variable is whether the radish seed is placed in the centrifuge or in normal earth conditions. The dependent variables are the overall growth of the plants which includes the stems diameter and length, which will be measured in millimeters in addition to the growth rate of the seeds (Millimeters/Day). A higher growth rate would signal more efficiency and a more optimal growth pattern. The constant factors of variables in this experiment are the other crucial conditions the plants are grown in, including light level, water amount, and temperature. Additionally, the time will be kept at a constant 3 day growing period. To analyze the significance of what G force level optimizes growth, we will use a 2 Sample T Test for means to figure out which condition has a beneficial effect on growth.

Experimental Design

Materials:

Slot Variable Speed Centrifuge	1 Gal Distilled water
(16) 10ml Test Tubes	100g Radish Seeds
(16) Test Tube Tops	Caliper
16+ Slot Test Tube Stand	Vernier Photogate
25ml Graduated Cylinder	Laptop
Forceps	Photogate Measuring Jig

Procedures:

1. Randomize all seed placement and choice. (see appendix A)
2. Set up 16 test tubes in the test tube stand.
3. Place 3 radish seeds in each test tube.
4. Measure 5 ml of distilled water using graduated cylinder and pour into each test tube.
5. Fit a test tube top on each test tube.
6. Place the photogate jig in the accelerometer.
7. Turn on the centrifuge to 1 speed level and measure velocity after 30 sec.
8. Use the G force equation to calculate G force (See appendix C).
9. Turn off the centrifuge and remove the Photogate Jig from the centrifuge.
10. Place 8 chosen seeded test tubes in the centrifuge.
11. Turn on the centrifuge to the previously measured setting.
12. Run the centrifuge for 24 hours while radishes sprout.
13. Turn off Centrifuge and remove test tubes.
14. Uncap varied G and remove seeds from the test tube.
15. Determine whether seed sprouted based on whether hypocotyl has matured from seed by more than 1mm using a caliper.

16. Place sprouted seeds on a flat surface and record unsprouted seeds.
17. Use a caliper to measure sprout from seed to top of sprout.
18. Record measurements collected.
19. Replace 3 seeds in each test tube and fill with 5ml water.
20. Place test tubes in Centrifuge and Turn on to level 1 speed.
21. Repeat steps 14-19 but for normal G seeds.
22. Run the test for an additional 24 hours then repeat steps 9-18.

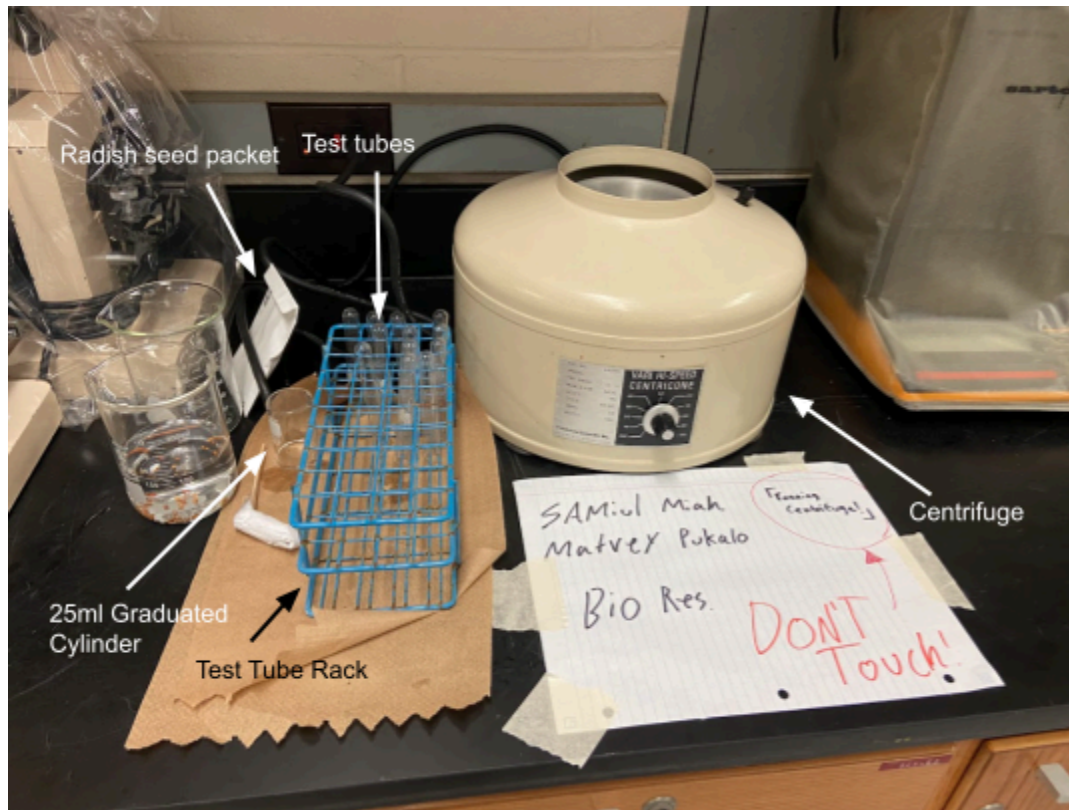
Diagrams:

Figure 8. Centrifuge and Test Tube

The figure above shows the test tube and centrifuge setup, the test tubes held in the test tube rack labeled above are experiencing regular earth gravity (1G) while those in the centrifuge would be experiencing 6Gs. Additionally, the 25 ml graduated cylinder that was used to measure the proper 5ml of water for each test tube is shown and labeled above. Radish seeds and test tubes are also labeled above

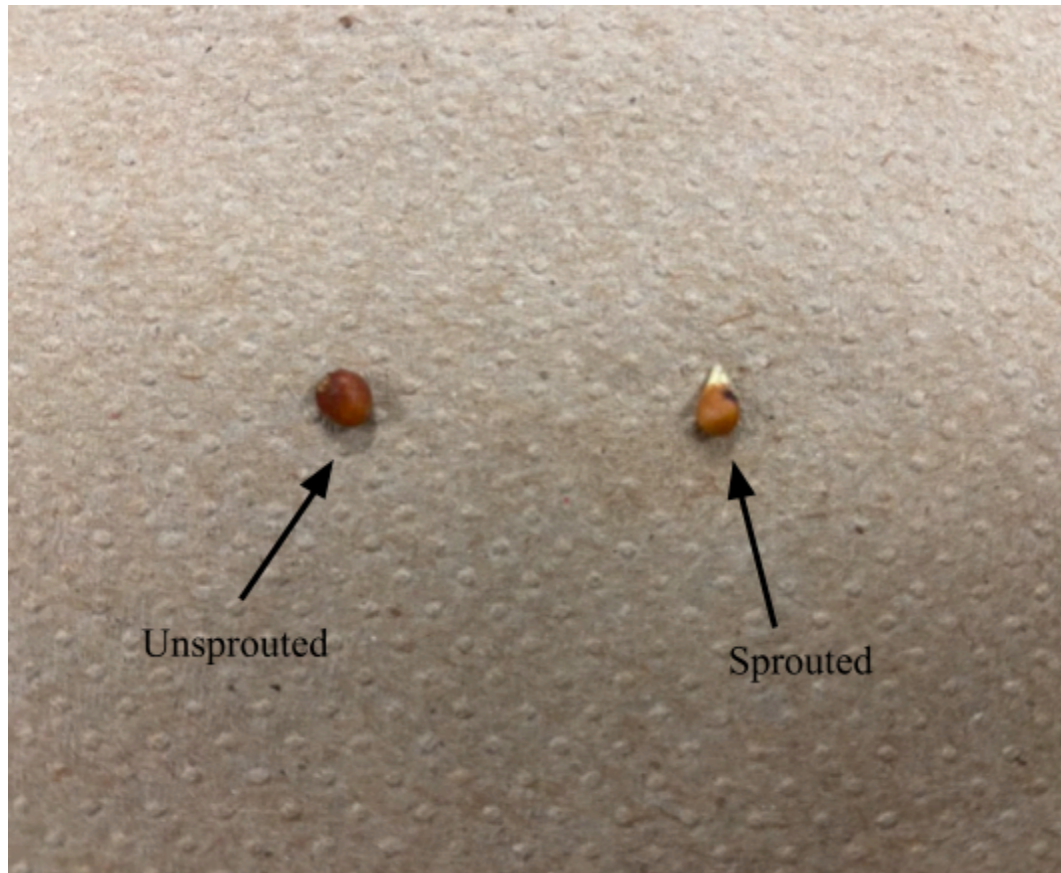


Figure 9. Sprouted vs Unsprouted Seeds

Figure 9 above displays 2 seeds, with the one on the left being considered unsprouted as the hypocotyl is less than 1mm long, whereas on the right it is considered to be sprouted as it is longer than 1mm and growing.

Data and Observations

Table 1

Data Tables for Radish Seeds Trial Set 1

Trial Number	Centrifuge Length(mm)	Non-Centrifuge Length (mm)	D2 Centrifuge Length (mm)	D2 Non-Centrifuge Length (mm)
1	0	0	0	0
2	0	0	6.9	0
3	0	0	2.99	0
4	0.5	0	4.19	0
5	2.57	0	6.25	0
6	1.29	0	1.56	0
7	4.31	0	0	0
8	3.5	0	3.52	0
9	3.4	0	2.55	0
10	0	0	3.18	0
11	2.98	1.75	5.71	0
12	2.19	0	0.9	0
13	0	0	3.19	1
14	0	0	1.63	0
15	1	0	2.73	0
16	0.85	0.5	4.7	0
17	0.85	0	5.59	0
18	0	0	6.12	0
19	0.58	0	0	0
20	0	0	0	0
21	0	0	4.7	1.65
22	2.84	0	0	2.72
23	0	0	6.37	0
24	0	0	4.6	0
AVERAGES	1.119166667	0.09375	3.224166667	0.22375

As shown in table 1 above, the data is given for the first batch of radish seeds which included trials 1-24 for the centrifuge and non centrifuge grown seeds. Data was collected at 24 and 48 hours. The lengths are recorded in millimeters (mm). The value 0 represents when a seed does

not sprout. This shows that the average sprout length for the centrifuge after 24 hours was 0.09mm, and 0.22 after 48 hours, while the average sprout length for normal was 1.12mm after 24 hours and 3.22 after 48.

Table 2
Data Tables for Radish Seeds Trial Set 2

Trial Number	Centrifuge Length(mm)	Non-Centrifuge Length (mm)	D2 Centrifuge Length (mm)	D2 Non-Centrifuge Length (mm)
25	0	0	0	0
26	0	0	0	2.09
27	0	0	0	1.18
28	0.5	0	0	0
29	0	0	0	0
30	0	0	1.36	0
31	0	0	0	0
32	0	0	0	0
33	0	0.3	1.56	3.79
34	0.1	0	0.97	0
35	0	0	2.72	0
36	0	0	0.84	0.1
37	0	0	1.44	0
38	0	0	1.69	0
39	0	0	0	0
40	0	0.5	3.56	0
41	1	0	1.81	0
42	0	0	0	0
43	0	0	0	0.98
44	0	0	0	0
45	1.3	0.6	0.98	0
46	0	0	0	0
47	0	0	0	1.56
48	0	0	0	0
AVERAGES	0.1208333333	0.05833333333	0.7054166667	0.4041666667

As shown in table 2, the data is given for the second batch of radish seeds which included trials 25-48 for the centrifuge and non centrifuge grown seeds. Data was collected at 24 and 48 hours. The lengths are recorded in millimeters (mm). The value 0 represents when a seed does not sprout. This shows that the average sprout length for the centrifuge after 24 hours was 0.05mm, and 0.4 after 48 hours, while the average sprout length for normal was 0.12mm after 24 hours and 0.71 after 48.

Table 3

Data Tables for Radish Seeds Trial Set 3

Trial Number	Centrifuge Length(mm)	Non-Centrifuge Length (mm)	D2 Centrifuge Length (mm)	D2 Non-Centrifuge Length (mm)
49	0	0	1.05	0
50	0	0	0	1.44
51	0	0	0	1.18
52	0	0	0	0
53	2.4	0	0	0
54	0.98	0	1.36	2.06
55	0	0	0	0
56	1.01	1.46	0	0
57	1.48	0.84	1.56	0.8
58	0	0	0.97	0
59	0	0	2.72	0
60	0	0	0.84	0.1
61	0	0	1.44	0
62	0	0	1.69	0
63	0	0	0	3.4
64	0	1.46	3.56	0
65	0	0	1.81	0
66	0	0	0	0
67	1.75	0	0	0.6
68	0.99	0	0	1
69	0	0	0.98	0
70	0	1.7	0	0

Trial Number	Centrifuge Length (mm)	Non-Centrifuge Length(mm)	D2 Centrifuge Length (mm)	D2 Non Centrifuge Length (mm)
71	0	2.09	0	1.56
72	0	0	0	0
AVERAGES	0.35875	0.31458	0.749166	0.50583

As shown in table 3 above, the data is given for the third batch of radish seeds which included trials 49-72 for the centrifuge and non centrifuge grown seeds. Data was collected at 24 and 48 hours. The lengths are recorded in millimeters (mm). The value 0 represents when a seed does not sprout. This shows that the average sprout length for the centrifuge after 24 hours was 0.35mm, and 0.75 after 48 hours, while the average sprout length for normal was 0.31mm after 24 hours and 0.5 after 48.

Table 4

Observation Table for Non-Centrifuge Grown Seeds for 24 Hours

Trial Number	Observations (Non-Centrifuge)
1	All seeds in the group did not sprout resulting in 0mm length.
2	All seeds in the group sprouted though all had dramatically different lengths.
3	All seeds in the group sprouted with similar length, though one did have much longer length and did not have seed casing around it exposing small leaves.
4	All but one seed in the group sprouted, two sprouted seeds had somewhat similar lengths.
5	All but one seed in the group did not sprout, the sprouted seed's shell casing was gone and leaves were exposed.
6	All but one seed in the group sprouted, sprouted seeds shared the same length.
7	All but one seed in the group did not sprout.
8	All but one seed in the group did not sprout.
9	Start of the second test batch, no seeds in the group sprouted.
10	All but one seed in the group did not sprout, the seed had yellow tint to stem and leaves.
11	No seeds in the group sprouted.
12	All but one seed in the group did not sprout, the sprouted seed was very small and the seed casing was very dark in color.
13	No seeds in the group sprouted.

Trial Number	Observations (Non-Centrifuge)
14	All but one seed in the group did not sprout, the sprouted seed's shell casing was gone with leaves exposed.
15	All but one seed in the group did not sprout, the sprouted seed's shell casing was gone with leaves exposed, other seeds very very dense and darkened.
16	No seeds in the group sprouted, all seeds were very soft and malleable.
17	Start of the 3rd test batch, no seeds in the group sprouted.
18	All but one seed in the group sprouted, a large difference in length between the two sprouted.
19	All but one seed did not sprout, two sprouts were somewhat similar in length.
20	No seeds in the group sprouted.
21	No seeds in the group sprouted.
22	No seeds in the group sprouted.
23	All but one seed in the group sprouted, the unsprouted seed was very soft and malleable, light colored.
24	No seeds in the group sprouted.

Table 4 above shows the observations that were made after the first 24 hours for the seeds grown in normal earth gravity conditions and not in the centrifuge. Each trial number consists of one test tube that all contain 3 seeds.

Table 5

Observation Table for Non-Centrifuge Grown Seeds for 48 Hours

Trial Number	Observations (Day 2 Non-Centrifuge)
1	Two of the three seeds in this group sprouted, one having the length of 6.9mm which is the longest recorded in all trials.
2	All three seeds in this group sprouted, with dramatically different lengths.
3	Two of the three seeds in this group sprouted, one was curved and one was relatively straight.
4	All three seeds in this group sprouted, with dramatically different lengths.
5	All three seeds in this group sprouted, with dramatically different lengths.
6	All three seeds in this group sprouted, with dramatically different lengths.
7	Two of the three seeds in this group sprouted, the sprouted seed was straight.
8	Two of the three seeds in this group sprouted, both varying significantly in length.

Trial Number	Observations (Day 2 Non-Centrifuge)
9	Start of the second test batch, no seeds in the group sprouted.
10	All but one seed in this group did not sprout, the sprouted seed being very straight.
11	All but one seed in this group did not sprout, the sprouted seed being very straight.
12	All three seeds in this group sprouted, two were similar in lengths.
13	Two of the seeds in this group sprouted and were somewhat similar in size and straight in shape.
14	Two of the seeds in this group sprouted, the lengths of the two were dramatically different.
15	All but one seed in this group did not sprout, the sprouted seed being very straight.
16	No seeds in the group sprouted.
17	Start of 3rd test batch, one seed sprouted, straight orientation
18	One seed in this group sprouted, the sprouted seed was very straight.
19	One seed in this group sprouted, the sprouted seed was very straight.
20	All three seeds in this group sprouted, all being different in size.
21	All but one seed in this group sprouted, one seed which sprouted was a brownish color.
22	All but one seed in this group sprouted, both sprouted being dramatically different in size.
23	One seed in this group sprouted, the sprouted seed was very straight .
24	No seeds in the group sprouted.

Table 5 above shows the observations that were made after the full testing period of 48 hours for the seeds grown in normal earth gravity conditions and not in the centrifuge. Each trial number consists of one test tube that all contain 3 seeds.

Table 6

Observation Table for Centrifuge Grown Seeds for 24 Hours

Trial Number	Observations (Centrifuge)
1	All seeds in the group did not sprout resulting in 0mm length. Water in the test tube evaporated over a 24 hour period.
2	All seeds in the group did not sprout resulting in 0mm length. Water in the test tube evaporated over a 24 hour period.
3	All seeds in the group did not sprout resulting in 0mm length. Water in the test tube evaporated over a 24 hour period.
4	All but one seed in the group did not sprout, the sprouted seed was curved in a horseshoe shape. Water in the test tube evaporated over a 24 hour period.
5	All seeds in the group did not sprout resulting in 0mm length. Water in the test tube evaporated over a 24 hour period.
6	All but one seed in the group did not sprout, the sprouted seed was curved in a horseshoe shape. Water in the test tube evaporated over a 24 hour period.
7	All seeds in the group did not sprout resulting in 0mm length. Water in the test tube evaporated over a 24 hour period.
8	All seeds in the group did not sprout resulting in 0mm length. Water in the test tube evaporated over a 24 hour period.
9	Start of trial set 2. All seeds in the group did not sprout resulting in 0mm length. Water in the test tube evaporated over a 24 hour period.
10	All seeds in the group did not sprout resulting in 0mm length. Water in the test tube evaporated over a 24 hour period.
11	All but one seed in the group did not sprout, the sprouted seed was curved in a horseshoe shape. Water in the test tube evaporated over a 24 hour period.
12	All seeds in the group did not sprout resulting in 0mm length. Water in the test tube evaporated over a 24 hour period.
13	All seeds in the group did not sprout resulting in 0mm length. Water in the test tube evaporated over a 24 hour period.
14	All but one seed in the group did not sprout, the sprouted seed was curved in a horseshoe shape and its shell disappeared. Water in the test tube evaporated over a 24 hour period.
15	All but one seed in the group did not sprout, the sprouted seed was curved in a horseshoe shape. Water in the test tube evaporated over a 24 hour period.
16	All seeds in the group did not sprout resulting in 0mm length. Water in the test tube evaporated over a 24 hour period.
17	Start of trial set 3. All seeds in the group did not sprout resulting in 0mm length. Water in the test tube evaporated over a 24 hour period.
18	All seeds in the group did not sprout resulting in 0mm length. Water in the test tube evaporated over a 24 hour period.

Trial Number	Observations (Centrifuge)
19	Two seeds in this group sprouted, both curved in a horseshoe shape. Water in the test tube evaporated over a 24 hour period.
20	All seeds in the group did not sprout resulting in 0mm length. Water in the test tube evaporated over a 24 hour period.
21	All seeds in the group did not sprout resulting in 0mm length. Water in the test tube evaporated over a 24 hour period.
22	All but one seed in the group did not sprout, the sprouted seed was curved in a horseshoe shape. Water in the test tube evaporated over a 24 hour period.
23	All seeds in the group did not sprout resulting in 0mm length. Water in the test tube evaporated over a 24 hour period.
24	Two seeds in this group sprouted, both curved in a horseshoe shape. Water in the test tube evaporated over a 24 hour period.

Table 6 above shows the observations that were made after the first 24 hours for the seeds grown in hypergravity conditions in the centrifuge. Each trial number consists of one test tube that all contain 3 seeds.

Table 7

Observation Table for Centrifuge Grown Seeds for 48 Hours

Trial Number	Observations (2 Day Centrifuge)
1	No seeds in the group sprouted.
2	No seeds in the group sprouted.
3	No seeds in the group sprouted.
4	No seeds in the group sprouted.
5	All but one seed in this group did not sprout, the sprouted seed was curved in a circular shape.
6	No seeds in the group sprouted.
7	All but one seed in this group did not sprout, the sprouted seed's stem was thin and curved like a horseshoe.
8	All but one seed in this group did not sprout, the sprouted seed's stem was thin and curved like a horseshoe.
9	Start of trial set 2. All but one seed in this group sprouted, all sprouted were curved in horseshoe patterns, unsprouted seed was black and hard.
10	No seeds in the group sprouted.

Trial Number	Observations (2 Day Centrifuge)
11	All but one seed in this group did not sprout, the sprouted seed was thin and bent at the end.
12	All but one seed in this group did not sprout, the sprouted seed was just barley sprouted.
13	No seeds in the group sprouted.
14	No seeds in the group sprouted.
15	All but one seed in this group did not sprout, the sprouted seed was bent in a "W" shape.
16	All but one seed in this group did not sprout, the sprouted seed was bent in a horseshoe shape.
17	Start of trial set 3. Two seeds in this group sprouted, the two seeds' stems were wrapped around each other.
18	All but one seed in this group did not sprout, the sprouted seed was bent in a horseshoe shape.
19	All but one seed in this group did not sprout, the sprouted seed was bent in a horseshoe shape.
20	All but one seed in this group did not sprout, the sprouted seed was just barely sprouted.
21	All but one seed in this group did not sprout, the sprouted seed was bent in a horseshoe shape.
22	No seeds in the group sprouted.
23	All but one seed in this group did not sprout, the sprouted seed was bent in a horseshoe shape.
24	All but one seed in this group did not sprout, the sprouted seed was bent in a U shape.

Table 7 above shows the observations that were made after the full trial sets of 48 total hours grown for the seeds grown in hypergravity conditions in the centrifuge. Each trial number consists of one test tube that all contain 3 seeds.

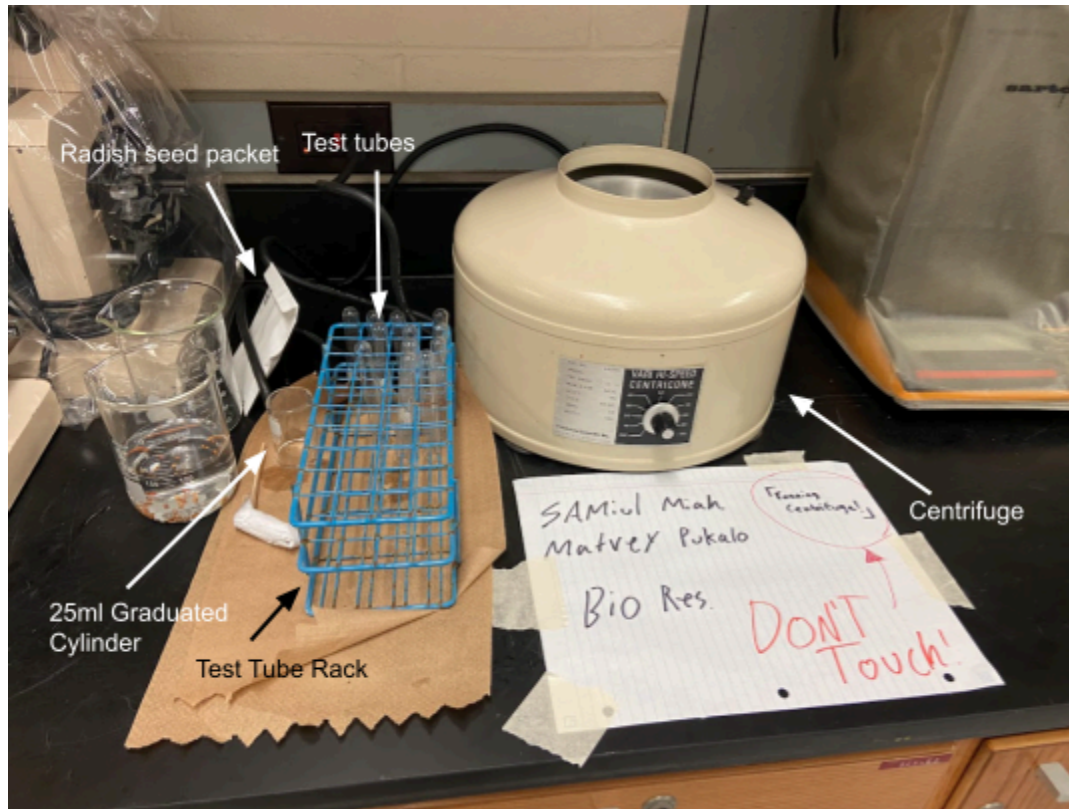


Figure 10. The Initial Set Up

Figure 10 above shows the setup for the experiment including the centrifuge, test tube rack, graduated cylinders, test tubes, and radish seeds. The experiment can be taken place with other centrifuges, however they will likely have different speeds and generate different g force levels, so that must be taken into account.

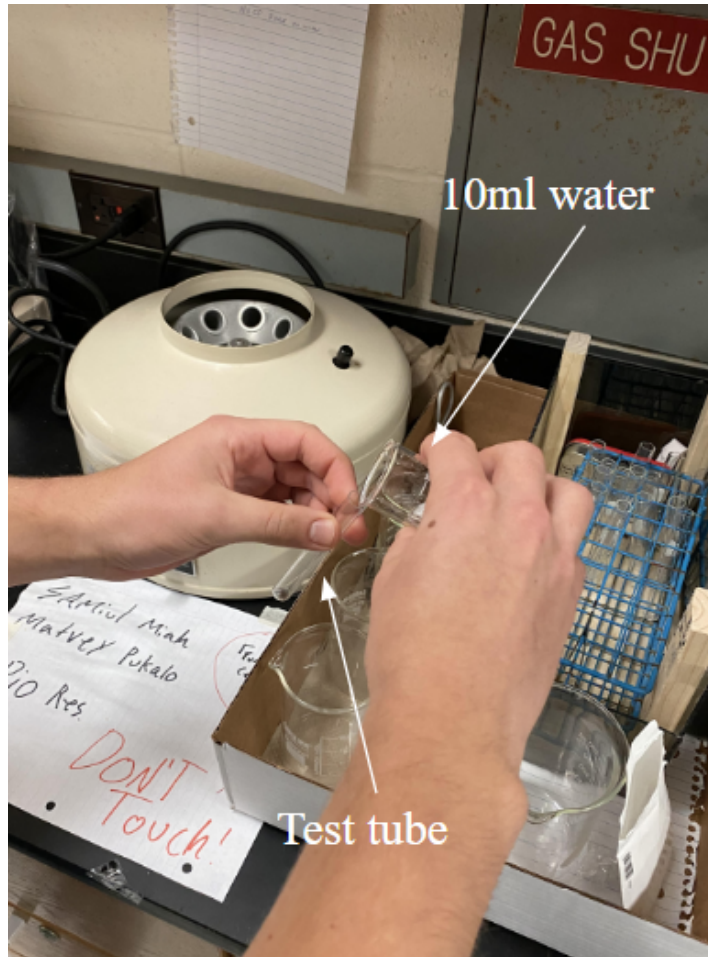


Figure 11. The First Step: Setting up Test Tubes

Figure 11 above shows the researchers pouring 10 ml of water into a test tube using a 25 ml graduated cylinder. The test tube has 3 radish seeds placed in it that were randomly selected. This ensured that the radish seeds were hydrated and aided their germination.

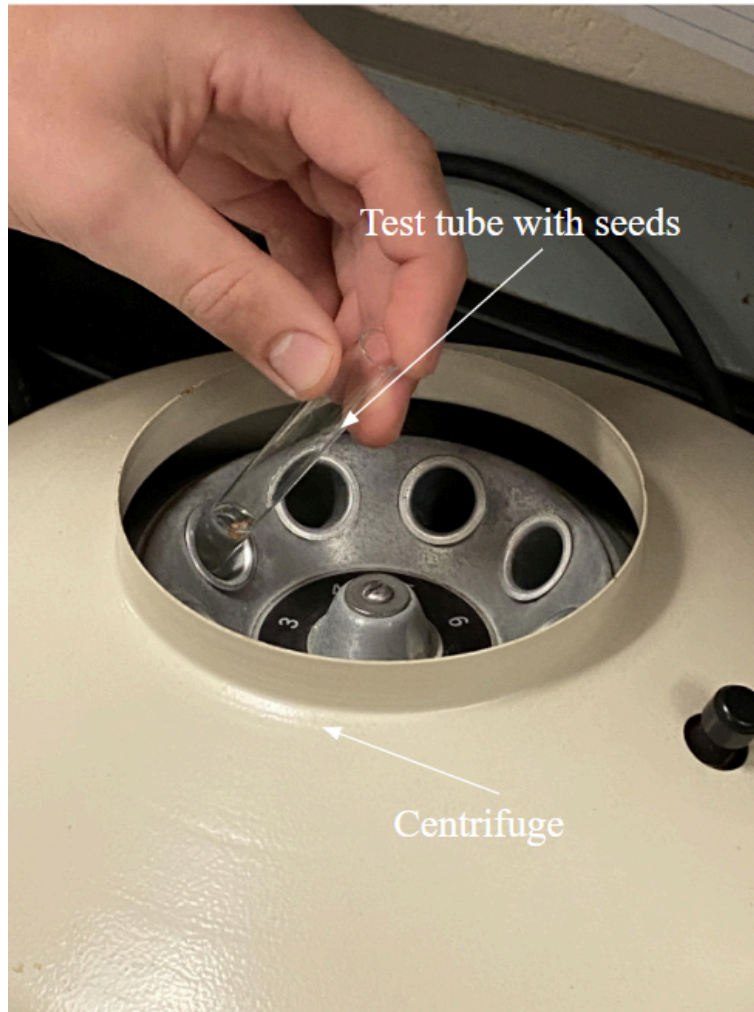


Figure 12. Step 2: Placing Test Tubes In Centrifuge

Figure 12 above shows the researchers placing a test tube with the contents of water and three seeds in the centrifuge. It is important to have all the test tubes be even and place test tubes in all slots or oriented opposite to each other to minimize vibrations and shaking.

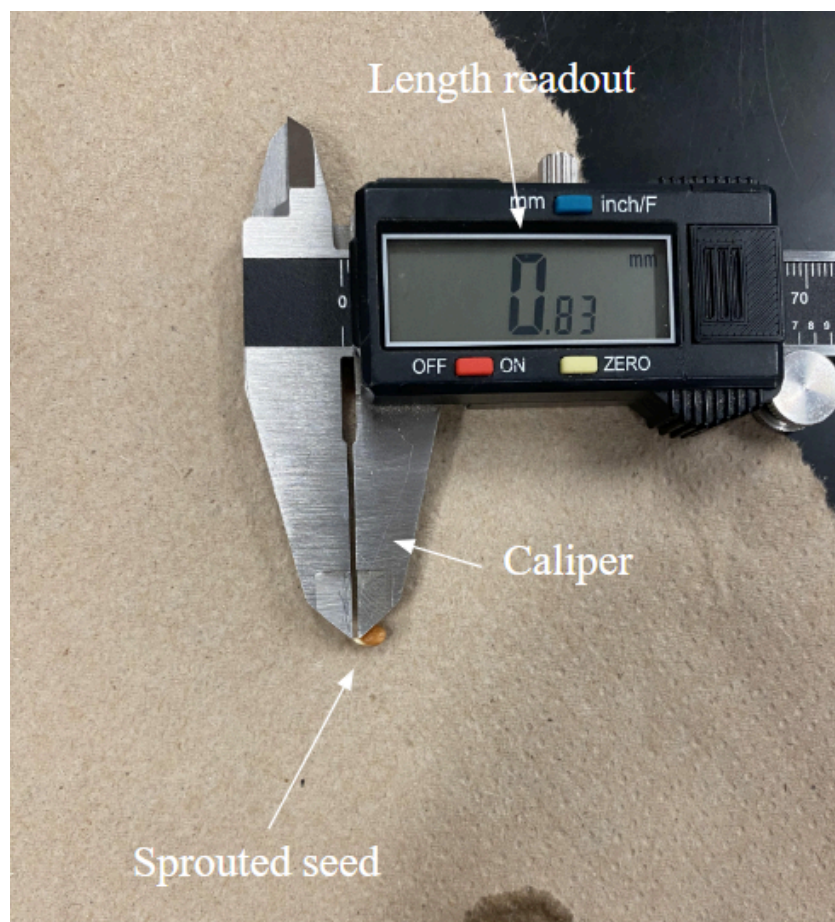


Figure 13. Step 3: Measuring sprouted seeds

Figure 13 above shows the researchers placing a seed that has sprouted on paper towel and measuring the length of the sprout from the top of the seeds shell to the end of the sprout.

Data Analysis and Interpretation

In the experiment, data on the sprouting percentage of sprouting and growth length was collected on radish seeds. These seeds were put through experiments where they were either placed in a centrifuge running at 6G's or in a test tube rack within a typical environment to grow 48 hours. Data was collected in two batches at intervals of 24 hours. Once this was done, their sprouting percentage was recorded, in addition to the length that they sprouted to in millimeters. These values for both sets of conditions were compared in 2 sample T-tests to determine if the seeds grown under hyper gravity had a significantly stunted growth rate when compared to normal gravity, or 1G. The experiments ran are comparative as it compares the average lengths and sprouting percentage of normal gravity to hyper gravity. The data collected is quantitative in the form of the sprouting length and qualitative in the form of if they sprouted or not. Precautions and changes were taken into account of the experiment to ensure the data was as accurate as possible. This includes the seeds that were chosen being randomized in their selection from the bag, to reduce variability if one part of the bag was contaminated or affected. Additionally, it was attempted to keep the environment as similar as possible within trials. This was accomplished through growing the two sets of seeds at the same location and at the same time. The test tubes were also washed out after use to remove any contaminants. There were a total of 144 trials run, with 72 being seeds grown in the centrifuge, and 72 being grown in normal earth conditions. 144 trials were run in an attempt to reduce variability and reveal any trends that occurred with the data.

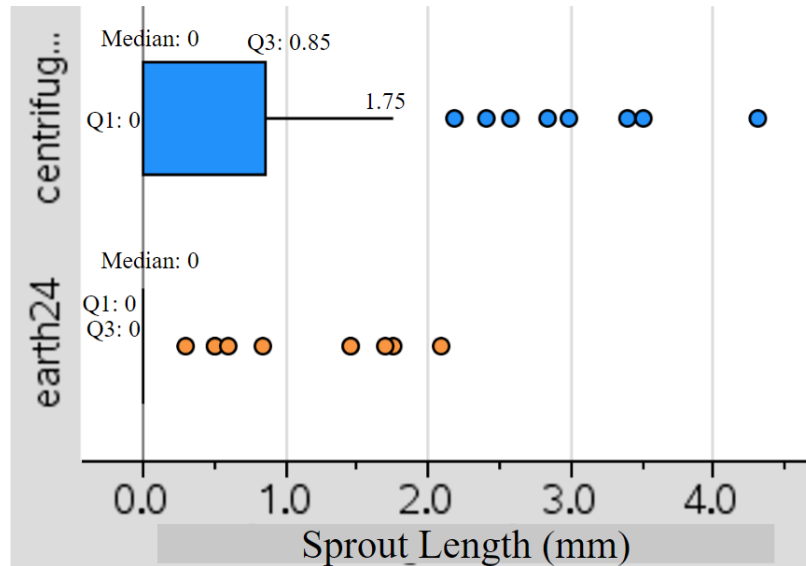


Figure 14. Box Plot of Values for Growth After 24 Hours

Figure 14 above compares 2 box plots, a box plot of the lengths of the seeds germinated in the centrifuge, shown in blue, to the seeds germinated normally, shown in orange. There is some overlap, as the box plot for the centrifuge grown seeds completely covers all of the data for the earth grown seeds. The medians for both is 0, which means that the majority of the seeds didn't sprout within the 24 hour period. Both plots also show a heavy right skew, showing that the majority of the trials did not sprout. Interestingly, the Q3 and tail for the earth gravity grown seeds were also 0, and all of the seeds that sprouted were counted as outliers. This plot shows already that there is a high probability that the seeds grown in the centrifuge sprouted more efficiently, as the Q3, tail, and outliers are all significantly higher than that of the earth gravity grown seeds.

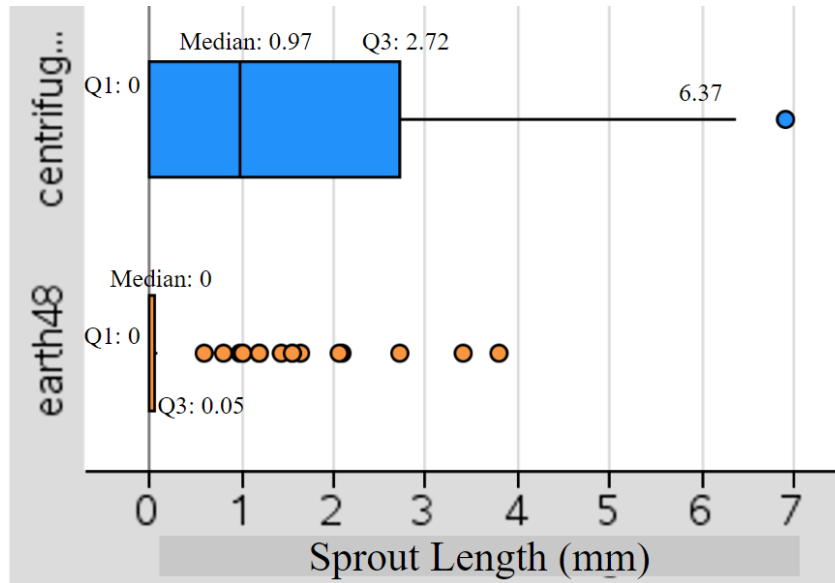


Figure 15. Box Plot of Values for Growth After 48 Hours

Figure 15 above shows a box plot of the data collected for the centrifuge grown seeds and the traditionally grown seeds over the full 48 hour test period. The first thing that it noticed is that the centrifuge grown seeds seemed to continue their trend of growing well, and the range for this also increased, with the median being 0.97. With the earth grown seeds, the median stayed the same at 0, however the Q3 increased to 0.05. The IQR (InterQuartile Range) for the centrifuge data covers almost all of the data from the earth grown, with the exception of two outliers. Both box plots are still right skewed, however, the centrifuge plot opened up more. This supports the idea from the box plot after 24 hours where the centrifuge seeds are shown to have a more efficient growth then that of the earth grown ones.

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Figure 16. Equation for 2 Sample t Test

Figure 16 above shows the equation for a 2 sample t test. The t represents the t value that will be collected from the calculation. The two $\bar{x}$ represent the sample means from the data collected, while the μ represents the population means. The n_s represent the standard deviations of both populations.

The appropriate statistical test chosen for the experiment is a two sample t test. This is due to the research comparing the means of 2 populations: length of seeds sprouted in centrifuge under hypergravity and under normal earth conditions. A 2 sample t test was used to definitively show whether the seeds sprouted faster under 1G, or normal earth conditions. Three assumptions need to be met to run a 2 sample t test. This includes normality, randomization, and the 10% condition. The 10% condition was met due to it being less than 10% of the population, or these seeds tested being less than 10% of all radish seeds on earth. Normality can be assumed due to the experiment containing a total of 144 trials, because after 30 or more trials, the experiment can be assumed to be normal as it will be more or less normal as you run more and more trials.

Randomization was conducted by randomizing the selection of each seed, and if they would be paced in the centrifuge or not. Because all of the conditions are met, it can be assumed that the results of a 2 sample t test would be reliable. For the equation and examples of a 2 sample t test, see appendix B.

$$H_o : \mu_C - \mu_E = 0$$

$$H_a : \mu_C - \mu_E < 0$$

Figure 17. Hypothesis Statements

Figure 17 above shows the statements for the null and alternative hypotheses. The null hypothesis, H_o , states that the growth rates of the hypergravity grown seeds weren't equal to that of the normal gravity grown seeds. The alternative hypothesis, H_a , states that the growth rate of the centrifuge grown seeds was less than the seeds under normal gravity conditions.

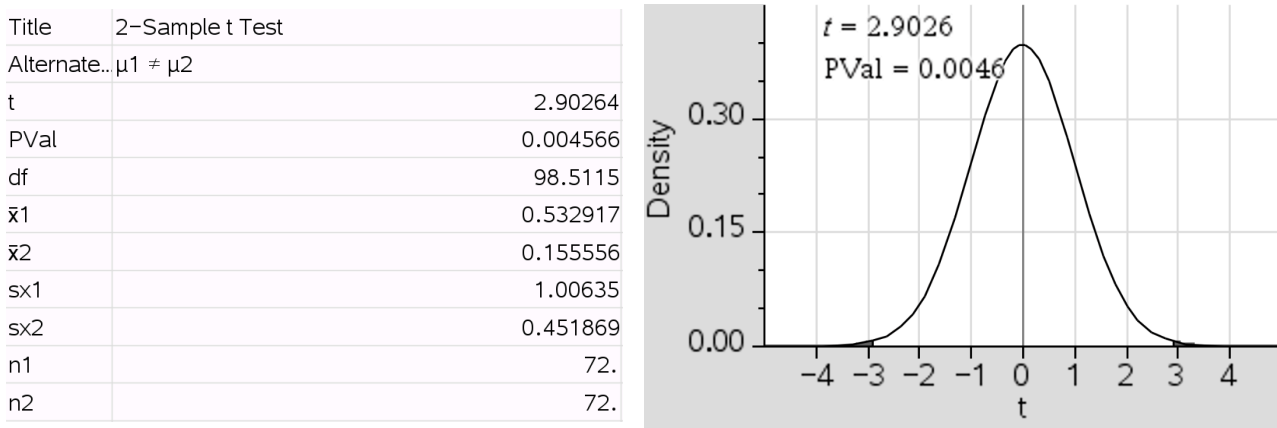


Figure 18. The 2-Sample t Test After 24 Hours

Figure 18 above shows the results collected from the 2 sample t test after the first interval of testing (24 Hours) and the probability graph that goes with it.

The null hypothesis is rejected because the p-value of 0.0046 is less than 0.05. There is evidence that the average growth of the seed grown in the centrifuge is not the same as the seeds grown in a normal earth environment. If the null hypothesis was true, meaning that the growth would be the same between the two tests, there would only be a 0.46% chance of getting results that are

not equal by chance alone. Because the probability of this happening is so low, the null hypothesis was rejected.

From this 2 sample t Test for the results after 24 hours and the box plots, it can be determined that there is a significant difference in the germination and growth of seeds, and that the seeds placed in the centrifuge grew more efficiently within 24 hours of the tests being conducted.

Title	2-Sample t Test
Alternate..	$\mu_1 \neq \mu_2$
t	4.84709
PVal	0.000003
df	191.583
$\bar{x}_1$	0.96875
$\bar{x}_2$	0.266736
s_{x1}	1.60329
s_{x2}	0.670856
n_1	144.
n_2	144.

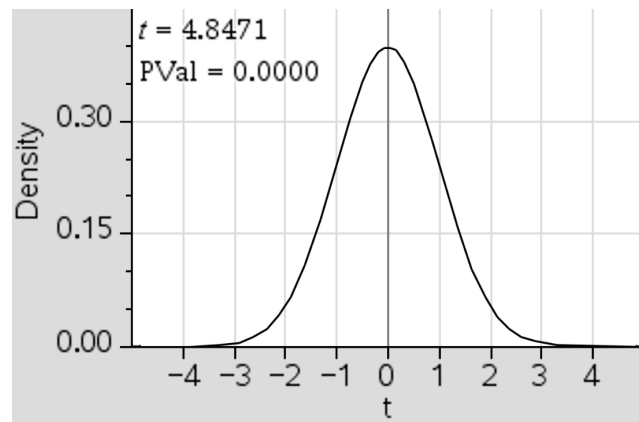


Figure 19. The 2-Sample t Test After 48 Hours

Figure 19 above shows the results collected from the 2 sample t test through the complete testing period of 48 hours and the probability plot that goes with it.

The null hypothesis was rejected because the p -value of basically 0 is less than 0.05. There is significant evidence that the germination and growth rates of seeds grown in a centrifuge and seeds grown in a normal earth environment are not the same over the full testing period of 48 hours. If the null hypothesis was true, meaning that the growth rates were the same across normal gravity and hypergravity, there would only be a $3.0 \times 10^{-6}\%$ chance of getting non equal results by chance alone. Because the probability of obtaining results that are unequal by chance alone happening is so low, the null hypothesis was rejected.

Due to the two sample t-tests showing that the growth rates from centrifuge hypergravity and the earth's normal gravity growth over a period of 48 hours being unequal, and the box plots showing that the centrifuge grown seeds had a significantly higher overall growth pattern including higher medians, tails, and IQR. This research shows that radish seeds grown in a hypergravity environment of 6 G's germinate and grow better than seeds grown in a normal earth environment.

Conclusion

The original hypothesis that seeds grow more efficiently in normal earth gravity (1G) than those grown at a higher G force (6Gs) was rejected. It was determined that the seeds grown in the hypergravity sprouted more efficiently than seeds grown in earth's gravity based on significant results from a two-sample t test comparing the mean lengths of growth of the seeds. A centrifuge was used as it could simulate a higher constant G force, which was set to 6Gs for testing. The seeds were then measured, and based off of the p-value and box plots, the hypergravity seeds grew significantly more. As can be seen in trial data tables previously, average length (mm) of centrifuge (higher G exposed) seeds were always greater than those growing at 1G, both in 24 hour exposure and 48 hour exposure. This is seen as in the first trial run where the 24 hour average length (mm) for centrifuge is 0.35875 while the non centrifuge was 0.31458.

These results revealed that the hypothesis that the radish seeds that were grown under normal 1G gravity conditions would yield the best overall growth and growth rate was rejected. The data showed that seeds under higher levels of Gs sprouted better.

The data collected does not support the original hypothesis that normal earth gravity grown seeds would grow more efficiently. In fact, the data shows almost the complete opposite; that the higher G forces germinated more efficiently. When the data was displayed in box plots, it showed various things. Firstly, over both growth periods of 24 hours and 48 hours, the plots showed that the majority of the seeds in both sets did not sprout, but the high G seeds had a wider range for the ones that did and consistently had higher data points. Over the 48 hour period, the earth gravity seeds still had a median of zero, the hypergravity seeds had a median length of 0.97mm, which already shows a significant difference in rates and evidence that higher gravity seemed to have an impact on the growth of the seeds. To provide more concrete

evidence, the data was put through two-sample t tests after 24 hours, and after the complete testing period. The two-sample t test after 24 hours yielded a p-value of 0.0046, which is significantly less than the alpha value of 0.05. Since the calculated p-value is much less than the alpha value, this shows that there was a significant difference between the two growth strategies, and proves that the hypergravity seeds grew more efficiently. After 48 hours, the test after 24 hours yielded a p-value of 3.0×10^{-6} , which is even further away from the alpha value of 0.05 and even more concretely supports that the hypergravity seeds sprouted better.

An experiment called “The Effect of Gravity on Plant Germination” was conducted that yielded different results to those collected throughout this research (Takakura). This study conducted research on how gravity would affect the growth of rice seeds. They placed rice seeds in test tubes under conditions with a constant temperature and light level then were put under varying G force levels. Their shoot and root growth was then measured every 12 hours for a 6 day period. Their results showed that microgravity had an effect on the shape and length of the shoot, however, under increased gravity (Hypergravity) there was no significant impact observed from increased gravity. This contrasts as it showed no effect compared to increased growth. One thing that was found to be similar however, is that the shape of the shoot of the seed differed. Under different gravity conditions, the diameter and orientation of the sprout would differ.

Gravity is one of the fundamental forces that act on life, and has an impact on everything in existence. Gravity can be defined as the force that attracts a body towards a physical body that has mass. It acts as the downwards force that keeps everything in place, and even more importantly opposes other forces that are trying to push outwards. With a higher gravity, more force and therefore energy is required to resist the gravity and for an object to stay or grow upward.

The process of the experimental design was helpful as it provided a way to revise the procedure and make it more efficient and reliable over time. The trials were run in the same room with the same ambient temperature, light levels, level of water and amount of seeds to minimize error. Even steps were taken, though outside factors may have had an impact on the data collected. Possibly the greatest outside factor was the heat produced by the centrifuge's motor. The electric motor from the centrifuge would run for 24-48 hours nonstop, causing the motor to create and release heat that was absorbed by the test tubes. This heating of the tubes created a slightly warmer environment for the seeds in the centrifuge. This was shown as the side of the centrifuge was slightly warm after trials and there seemed to be more evaporation occurring in the tubes placed in the centrifuge. The optimal germination temperature for radish seeds is between 55-75 degrees Fahrenheit. When the temperatures of the water in the test tubes were measured, the centrifuge seeds read 78 degrees which is outside of the stated optimal range, while the room temp seeds were 69 degrees, which is inside the optimal range. Another factor which could have possibly affected data collected could be how precise the centrifuge was. The centrifuge used in this experiment was considerably aged and slight movements of the dial would result in large changes in rpm. Even a slight difference in rpm may have resulted in a noteworthy change in g-force.

Future improvements for more reducing error in data collection with this experiment would include calculating the amount of water lost in the centrifuge and adding a slightly more water in adjustment for the amount that would be lost before measuring. The amount of water in both the centrifuge group and room temp group would then be the same at the time of data collection, this would also help account for water evaporated from the centrifuge. Water evaporating from the test tubes could also be solved by covering the open top of the tube. The

use of a more recent/ more accurate centrifuge would also help in maintaining consistency between data sets. Preferably one which a consistent level of rpm could be set would aid in reducing error.

The research process and results can be used in the future for other trials and further research. The results of this experiment can be used in similar future experiments which deal with the germination of plants under the effects of g-forces not natural to that of earth. By identifying if there are other significant factors which affect seed germination and exploring those in depth the results of this experiment could be used in tandem to expand our understanding on this topic. The takeaways of this experiment can also be applied to stages of plant growth past germination, such as how higher g-forces affect stem growth in the aspect of the stems thickness/ overall health. The results could also serve as a basis for similar testing done on other types of seeds which are not radish. Though the further applications do not just apply to plant life, as it could also be applied to other forms of life. The effects of germination of plants under higher g-forces could be analyzed more in depth to understand the exact effect being had on a cellular level, which would then allow researchers to understand how animal cells would grow under the previously mentioned conditions. This would allow for a greater understanding/ prediction of how growth/germination of plants and animals would occur on planets/ extraterrestrial situations with heavier gravity than on earth.

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Appendix A: 2 Sample t test

A 2 sample t test is the statistical analysis that is used in the research. A t test can compare 2 values of means to each other and find the probability of them being the same, equal, or greater/less. The equation is run by subtracting the means, $\overline{x}$, of both subjects, then dividing that by the square root of the sum of the standard deviations, s , over their sample size, n , for each subject respectively.

$$t = \frac{(\overline{x}_1 - \overline{x}_2)}{\sqrt{\left(\frac{s_1^2}{n_1}\right) + \left(\frac{s_2^2}{n_2}\right)}}$$

Figure 20 below shows the sample calculation for a 2 sample t test.

$$t = \frac{(23.2 - 100.7)}{\sqrt{\left(\frac{0.7^2}{30}\right) + \left(\frac{3.4^2}{30}\right)}}$$

$$t = \frac{-77.5}{\sqrt{0.407}}$$

$$t = -121.48$$

Figure 20. Sample Calculation for 2 Sample t test

Appendix B: Randomizing Trials with TI-Nspire Graphing Calculator

1. Find the number of trials being tested and allocate a number for each seed. In the calculator, Type in randInt, and in the parentheses type in 1 and then the number of seeds tested, then the final number to represent how many numbers will be generated. An example is shown below.

 $\text{randInt}(1,8,8)$ $\{ 8,2,3,1,8,1,1,5 \}$

2. If any numbers are repeated like the 8 and 1's. Redo the calculation and disregard numbers that previously appeared.
3. Repeat steps 1-3 for all trial runs.

Appendix C: Calculating G Force From Centrifuge

The G force that was produced by the centrifuge was not given in the experiment and needed to be calculated. This was calculated through first collecting the RPM (Rotations Per Minute) using a photogate that measured the rotations from the centrifuge. From there, that RPM value was used to calculate the velocity of the movement. This was later combined with the radius of the centrifuge in meters to determine the centripetal acceleration. Finally, the lateral G force was calculated by dividing the acceleration by 9.8, or the acceleration of gravity.

$$v = RPM \times \frac{2\pi r}{60}$$

$$a = \frac{v^2}{r}$$

$$G = \frac{a}{9.8}$$

Figure 21 shows a sample calculation for the G force of the centrifuge.

$$v = 637 \times \frac{2\pi(0.145)}{60} = 9.67$$

$$a = \frac{9.67^2}{0.145} = 64.89$$

$$G = \frac{64.89}{9.8} = 6.58$$

Figure 21. Sample calculation for G force.

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